

# Exact Rank of Complex Hankel Tensors

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12 September 2026

## Abstract

Let  $H_{i_1 \dots i_m} = h_{i_1 + \dots + i_m}$  for  $0 \leq i_j \leq n-1$ , over  $\mathbb{C}$ . Write  $D = m(n-1)$ , and let  $r$  be the rank of the middle Hankel catalecticant. For  $H \neq 0$ , choose a homogeneous binary apolar polynomial of smallest degree  $r$ , and let  $s$  be its number of distinct projective zeros. We prove the exact formula

$$R(H) = R_{\text{sym}}(H) = \min\{D - r + 2, (m-1)r - (m-2)s\}.$$

When the smallest-degree apolar polynomial is not unique, the first term is  $r$ , so the formula is independent of this choice. This proves the universal exact-rank equality requested in TR-14, including exceptional and mixed-multiplicity tensors. The lower bound uses a contextual product-space inequality: an annihilating Frobenius functional excludes nilpotents from the auxiliary stabilizer algebras, leaving finitely many support partitions. The upper bounds are explicit symmetric interpolation constructions. We also prove that all three border ranks equal  $r$ , and that the maximum ordinary and symmetric Hankel ranks are  $(m-1)(n-1) + 1$ .

## 1 Statement and conventions

Throughout, the field is  $\mathbb{C}$ ,  $m \geq 3$ ,  $n \geq 2$ ,  $q = n-1$ , and  $D = mq$ . Tensor indices are zero-based. Thus the tensor considered here is exactly the one in TR-14 after subtracting one from each index. Ordinary rank permits arbitrary decomposable summands; symmetric rank permits only scalar multiples of  $v^{\otimes m}$ . The question imposes neither a genericity assumption nor a Vandermonde restriction.<sup>1</sup>

The problem originates in Nie–Ye, Conjecture 6.3.<sup>2</sup> The generic odd-order question is distinct: the repository’s TR-13 manuscript uses a compressed Koszul flattening and explicitly leaves exceptional tensors outside its conclusion.<sup>3</sup> The exact lower bound below does not follow by specialization from a generic rank statement.

Define the homogeneous moment functional

$$L_D(X^{D-j}Y^j) = h_j, \quad 0 \leq j \leq D.$$

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<sup>1</sup>Open Problems in Numerical Linear Algebra, TR-14, “Comon’s exact-rank conjecture for Hankel tensors,” accessed September 12, 2026. <https://github.com/ajt60gaibb/OpenProblemsInNLA/tree/main/tensor-computations/TR-14>.

<sup>2</sup>J. Nie and K. Ye, *Hankel Tensor Decompositions and Ranks*, SIAM J. Matrix Anal. Appl. 40 (2019), 486–516. <https://doi.org/10.1137/18M1168285>.

<sup>3</sup>M. J. Colbrook, repository manuscript for TR-13, September 11, 2026. <https://github.com/ajt60gaibb/OpenProblemsInNLA/tree/main/tensor-computations/TR-13>.

Equivalently, in the affine chart  $X = 1$ , set  $L(t^j) = h_j$  for  $j \leq D$ . Identifying a mode with  $\mathbb{C}[t]_{\leq q}$ , the tensor is the multilinear form

$$H(p_1, \dots, p_m) = L(p_1 \cdots p_m). \quad (1)$$

Using dual spaces for this interpretation does not change either rank. Vandermonde rank  $R_V$  additionally requires each symmetric factor to lie on  $\{(1, t, \dots, t^q) : t \in \mathbb{C}\} \cup \{(0, \dots, 0, 1)\}$ , up to scaling. Border rank, with any of these three restrictions, is the least rank bound whose complex algebraic closure contains the tensor; equivalently, it permits limits of tensors with that rank bound. The associated binary form is

$$F(x, y) = \sum_{j=0}^D \binom{D}{j} h_j x^{D-j} y^j.$$

For  $0 \leq d \leq D$ , define the apolar kernel

$$I_d = \left\{ \sum_{i=0}^d g_i X^{d-i} Y^i : \sum_{i=0}^d g_i h_{i+j} = 0 \quad (0 \leq j \leq D-d) \right\}. \quad (2)$$

This is the usual degree- $d$  apolar kernel of  $F$ . For  $d > D$ , put  $I_d = \mathbb{C}[X, Y]_d$ . The spaces  $I_d$  form a homogeneous ideal, directly from the displayed equations.

For  $H \neq 0$ , let  $r$  be the smallest degree for which  $I_r \neq 0$  and choose  $0 \neq G \in I_r$ . Let  $s$  be the number of distinct zeros of  $G$  in  $\mathbb{P}^1$ , counted without multiplicity. Section 2 proves

$$r = \text{rank } C_h, \quad C_h = (h_{i+j})_{\substack{0 \leq i \leq \lfloor D/2 \rfloor \\ 0 \leq j \leq \lceil D/2 \rceil}}. \quad (3)$$

Thus the invariants in the next theorem are computable by exact linear algebra and a univariate greatest common divisor.

**Theorem 1.1** (Exact rank formula; resolution of TR-14). *For every nonzero complex Hankel tensor of order  $m \geq 3$  and dimension  $n \geq 2$ ,*

$$\boxed{R(H) = R_{\text{sym}}(H) = \min\{D - r + 2, (m-1)r - (m-2)s\}. \quad (4)$$

*If  $I_r$  has dimension greater than one, then  $D = 2r - 2$  and both ranks are  $r$ , regardless of the choice of  $G$ . The zero tensor has both ranks zero.*

The proof has three ingredients. The minimal apolar polynomial gives a nondegenerate finite moment algebra. Two symmetric constructions give the upper bounds in (4). A product-space argument gives a matching lower bound for an arbitrary minimum-length *ordinary* decomposition. That last argument, not a restriction of a full-algebra rank bound, is the part that treats  $r > n$  and multiple repeated support points.

## 2 The finite moment algebra

A linear change in  $(X, Y)$  acts invertibly on each degree- $q$  mode in (1). It therefore preserves ordinary and symmetric tensor rank, and preserves the projective multiplicities of  $G$ . Choose such a change so that  $G$  has no zero at infinity. After scaling,

$$g(t) = G(1, t) = t^r + g_{r-1}t^{r-1} + \cdots + g_0$$

is monic of degree  $r$ . Write

$$g(t) = \prod_{a=1}^s (t - \alpha_a)^{\ell_a}, \quad \ell_a \geq 1, \quad \sum_a \ell_a = r. \quad (5)$$

**Lemma 2.1** (Moment algebra). *The algebra  $A = \mathbb{C}[t]/(g)$  has a functional  $\lambda : A \rightarrow \mathbb{C}$  such that*

$$\lambda([t^j]) = h_j \quad (0 \leq j \leq D). \quad (6)$$

*The pairing  $(a, b) \mapsto \lambda(ab)$  is nondegenerate. Moreover,  $r \leq \lfloor D/2 \rfloor + 1$ , and (3) holds.*

*Proof.* At degree  $d = \lfloor D/2 \rfloor + 1$ , the linear map in (2) has  $d + 1$  columns and  $D - d + 1$  rows, with more columns than rows. This proves the bound on  $r$ . Define  $\lambda$  on the basis  $1, t, \dots, t^{r-1}$  by the first  $r$  moments. The recurrence

$$h_{j+r} + g_{r-1}h_{j+r-1} + \dots + g_0h_j = 0 \quad (0 \leq j \leq D - r)$$

then proves (6) successively through degree  $D$ .

The radical of the pairing,  $J = \{a \in A : \lambda(aA) = 0\}$ , is an ideal. If  $J \neq 0$ , then  $A/J \simeq \mathbb{C}[t]/(g')$  for a monic proper divisor  $g'$  of  $g$ . The functional descends to this quotient. Equation (6) implies that  $g'$  gives a recurrence for all the required moments, so its homogenization is a nonzero element of  $I_{\deg g'}$ . This contradicts the minimality of  $r$ . Hence the pairing is nondegenerate.

Both polynomial spaces of degrees  $\lfloor D/2 \rfloor$  and  $\lceil D/2 \rceil$  surject onto  $A$ , because both degrees are at least  $r - 1$ . The middle catalecticant is the pullback of this nondegenerate pairing through these two surjections. Its rank is therefore  $r$ . The rank is unchanged on returning to the original binary coordinates.  $\square$

A functional with the nondegeneracy property in Lemma 2.1 is called *Frobenius*. The Chinese remainder decomposition is

$$A \simeq \prod_{a=1}^s A_a, \quad A_a = \mathbb{C}[z_a]/(z_a^{\ell_a}), \quad z_a = t - \alpha_a. \quad (7)$$

Every restriction  $\lambda_a$  is Frobenius. For a nonempty subset  $J \subseteq \{1, \dots, s\}$ , let  $A_J = \prod_{a \in J} A_a$  and  $r_J = \sum_{a \in J} \ell_a$ . Polynomial reduction gives

$$\dim \text{im}(\mathbb{C}[t]_{\leq q} \longrightarrow A_J) = \min\{n, r_J\}. \quad (8)$$

Also,

$$\text{span}_{\mathbb{C}}\{[p_1] \cdots [p_{m-1}] : \deg p_i \leq q\} = A, \quad (9)$$

because  $(m - 1)q \geq r - 1$  and products span every polynomial of degree at most  $(m - 1)q$ .

**Lemma 2.2** (Choice of the minimal apolar polynomial). *If  $D \geq 2r - 1$ , then  $I_r$  is one-dimensional. If  $D = 2r - 2$ , then  $\dim I_r = 2$ .*

*Proof.* In the first case, polynomials of degree at most  $D - r$  surject onto  $A$ . For  $b \in I_r$ , the equations  $\lambda([b][t^j]) = 0$  for  $j \leq D - r$  and nondegeneracy give  $[b] = 0$ . Since  $\deg b \leq r$ ,  $b$  is a multiple of  $g$ . In the second case, polynomials of degree at most  $r - 2$  give  $r - 1$  independent functionals on  $A$ . Reduction of polynomials of degree at most  $r$  onto  $A$  has one-dimensional kernel, so the apolar kernel has dimension two.  $\square$

### 3 Two symmetric upper bounds

**Proposition 3.1** (Local interpolation upper bound). *With (5),*

$$R_{\text{sym}}(H) \leq \sum_{a=1}^s ((m-1)(\ell_a - 1) + 1) = (m-1)r - (m-2)s. \quad (10)$$

*Proof.* In  $A_a = \mathbb{C}[z]/(z^\ell)$ , write

$$\lambda_a(f) = [z^{\ell-1}] u(z)f(z), \quad u(0) \neq 0.$$

Such  $u$  exists uniquely, and its constant coefficient is nonzero because  $\lambda_a$  is Frobenius. Choose an  $m$ th root  $w$  of  $u$  modulo  $z^\ell$ ; this exists by the finite binomial expansion, after choosing an  $m$ th root of  $u(0)$ . Set  $N = (m-1)(\ell-1) + 1$ . For polynomials  $f_i$  of degree at most  $\ell-1$ ,

$$[z^{\ell-1}] \prod_{i=1}^m f_i(z) = \frac{1}{N} \sum_{\zeta^N=1} \zeta^{-(\ell-1)} \prod_{i=1}^m f_i(\zeta). \quad (11)$$

Indeed, among integers between 0 and  $m(\ell-1)$ , the only one congruent to  $\ell-1$  modulo  $N$  is  $\ell-1$  itself. Apply (11) to  $f_i = w(z)p_i(\alpha_a + z) \bmod z^\ell$ . Each summand uses the same linear functional of  $p_i$  in every mode, hence is symmetric rank one. Summing the local decompositions gives (10). The case  $\ell = 1$  has  $N = 1$  and is included.  $\square$

**Proposition 3.2** (Binary interpolation upper bound). *Every nonzero tensor under consideration satisfies*

$$R_{\text{sym}}(H) \leq R_V(H) \leq D - r + 2. \quad (12)$$

*Proof.* Put  $v = D - r + 2 \geq r$ . Let  $\tau : A \rightarrow \mathbb{C}$  take the coefficient of  $t^{r-1}$  in the remainder modulo  $g$ . This is Frobenius: for a nonzero polynomial of degree  $d < r$ , multiplication by  $t^{r-1-d}$  produces a nonzero top coefficient. Thus  $\lambda(a) = \tau(ua)$  for a unique unit  $u \in A$ . Choose the degree-at-most- $r-1$  representative  $b(t)$  of  $u^{-1}$ . For  $0 \leq j \leq r-2$ ,

$$L(t^j b(t)) = \lambda([t^j b]) = \tau([t^j]) = 0.$$

Consequently the degree- $v$  homogenization  $B(X, Y) = X^v b(Y/X)$  is apolar. It is coprime to  $G$ : its residue modulo  $g$  is a unit, and  $G$  has no zero at infinity.

Choose a homogeneous  $Q$  of degree  $v - r$  so that  $GQ$  and  $B$  are coprime and  $Q$  does not vanish at infinity. The pencil  $B + cGQ$  has a squarefree member with no root at infinity. To see this directly, a repeated finite zero would be a critical point of the nonconstant rational function  $B/(GQ)$ ; in characteristic zero there are only finitely many critical values. Exclude these finitely many values and  $c = 0$ .

Let the resulting affine polynomial have distinct roots  $\beta_1, \dots, \beta_v$ . Solve the invertible Vandermonde system for weights  $\gamma_a$  matching  $h_0, \dots, h_{v-1}$ . Apolarity gives the recurrence which then matches every remaining moment through degree  $D$ . Hence

$$H = \sum_{a=1}^v \gamma_a (1, \beta_a, \dots, \beta_a^q)^{\otimes m}.$$

Zero weights may be discarded. For  $r = 1$ , the same argument uses an empty set of conditions  $0 \leq j \leq -1$ ; alternatively Lemma 2.1 immediately gives rank one.  $\square$

**Proposition 3.3** (Vandermonde rank dichotomy). *The exact Vandermonde rank is*

$$R_V(H) = \begin{cases} r, & G \text{ is squarefree,} \\ D - r + 2, & G \text{ is not squarefree.} \end{cases}$$

*This is independent of the choice of  $G$  in the balanced case  $D = 2r - 2$ .*

*Proof.* A Vandermonde decomposition with  $d$  distinct projective nodes gives a squarefree polynomial in  $I_d$ : take the product of the linear forms vanishing at the nodes. Conversely, a squarefree apolar polynomial yields a decomposition on its nodes by the Vandermonde recurrence construction, after moving infinity away from the nodes. Thus  $R_V(H)$  is the smallest squarefree apolar degree.

For  $r \leq d < D - r + 2$ , every  $b \in I_d$  satisfies  $\lambda([b][t^j]) = 0$  for  $j \leq D - d$ , and  $D - d \geq r - 1$ . Nondegeneracy forces  $[b] = 0$ , so in homogeneous notation  $I_d = G\mathbb{C}[X, Y]_{d-r}$ . If  $G$  is not squarefree, none of these polynomials is squarefree; Proposition 3.2 attains degree  $D - r + 2$ . If  $G$  is squarefree, degree  $r$  is already attainable and minimal by definition of  $r$ . When  $D = 2r - 2$ , both alternatives give  $r$ .  $\square$

The familiar binary interpolation construction is thus sufficient for the first bound; no genericity is used. The local construction supplies the potentially smaller second bound even when  $r > n$ .

## 4 A product-space inequality with a Frobenius constraint

For subspaces  $U, V$  of a commutative algebra,  $UV$  denotes the linear span of all products  $uv$ . Empty products of subspaces mean  $\mathbb{C}1$ . For a subspace  $W$ , define

$$\text{Stab}(W) = \{b : bW \subseteq W\}.$$

It is a unital subalgebra. The following transformation lemma is the commutative finite-dimensional specialization of Beck–Lecouvey, Lemma 4.2.<sup>4</sup>

**Lemma 4.1** (Linear transformation lemma). *Let  $B$  be a finite-dimensional commutative complex algebra. If  $U, V \subseteq B$  contain units, then for every unit  $a \in U$  there are a unital subalgebra  $K_a \subseteq B$  and a subspace  $T_a$  such that*

$$aV \subseteq T_a \subseteq UV, \quad K_a T_a = T_a, \quad \dim T_a + \dim K_a \geq \dim U + \dim V. \quad (13)$$

*Proof.* Scaling reduces to  $a = 1 \in U$  and  $1 \in V$ . Induct on  $\dim U$ . For a unit  $e \in V$ , put

$$U_e = U \cap Ve^{-1}, \quad V_e = V + Ue.$$

Then  $1 \in U_e \cap V_e$ ,  $U_e V_e \subseteq UV$ , and  $\dim U_e + \dim V_e = \dim U + \dim V$ . If some  $U_e \neq U$ , apply induction to  $(U_e, V_e)$ . Otherwise  $Ue \subseteq V$  for every unit  $e \in V$ . Units span  $V$  (the nonunits form finitely many proper residue hyperplanes), so  $UV = V$ . The algebra  $K = \mathbb{C}[U]$  satisfies  $KV = V$  and  $\dim K \geq \dim U$ ; take  $T = V$ . This also covers the initial case  $\dim U = 1$ . Undoing the unit scalings gives (13).  $\square$

<sup>4</sup>V. Beck and C. Lecouvey, *Additive combinatorics methods in associative algebras*, Confluentes Math. 9 (2017), no. 1, 3–27, Lemma 4.2. <https://doi.org/10.5802/cml.34>.

A general product-space Kneser inequality cannot simply be invoked in an arbitrary nonreduced algebra. The next lemma proves exactly the contextual version needed here and supplies the missing finiteness argument.

**Lemma 4.2** (Contextual product inequality). *Let  $B = A \times \mathbb{C}^R$ , where  $A$  is a finite-dimensional commutative complex algebra. Let  $\Lambda : B \rightarrow \mathbb{C}$  restrict to a Frobenius functional on  $A$ . Suppose  $U_1, \dots, U_m \subseteq B$  contain 1, and set  $W = U_1 \cdots U_m$ . Assume*

$$\Lambda(W) = 0, \quad \pi_A \left( \prod_{k \neq i} U_k \right) = A \quad \text{for every } i. \quad (14)$$

For  $W_j = U_1 \cdots U_j$  and  $j \geq 2$ ,

$$\dim W_j \geq \dim W_{j-1} + \dim U_j - \dim \text{Stab}(W_j). \quad (15)$$

Moreover,  $\text{Stab}(W)$  is reduced.

*Proof.* Fix  $j$ , put  $U = U_j$ ,  $V = W_{j-1}$ , and  $Q = U_{j+1} \cdots U_m$ . Apply Lemma 4.1 to a unit  $a \in U$ , obtaining  $K_a, T_a$ . If  $z \in K_a$  is nilpotent, its  $\mathbb{C}^R$  component is zero. Since  $zT_a \subseteq T_a$ ,

$$z a V Q \subseteq T_a Q \subseteq UVQ = W.$$

Consequently

$$\Lambda_A(z_A a_A \pi_A(VQ)) = 0.$$

Here  $\pi_A(VQ) = A$  by (14), and  $a_A$  is a unit. Nondegeneracy of the Frobenius pairing forces  $z_A = 0$ , hence  $z = 0$ . Thus *every auxiliary algebra  $K_a$  is reduced*.

There are only finitely many reduced unital subalgebras of  $B$ . Indeed, a reduced finite-dimensional complex algebra is a product of copies of  $\mathbb{C}$ . Its primitive idempotents map to idempotents of  $B$ . In each local factor of  $B$  an idempotent is either 0 or 1. Such subalgebras are therefore specified by partitions of the finite set of local factors of  $B$ .

Let  $d = \dim U$  and choose a basis  $x_1 = 1, x_2, \dots, x_d$  of  $U$ . For  $c \in \mathbb{C}$ , set

$$a(c) = x_1 + cx_2 + \cdots + c^{d-1}x_d.$$

All but finitely many  $a(c)$  are units: each residue coordinate is a polynomial with constant term 1. Finiteness of the possible  $K_{a(c)}$  gives  $d$  distinct values  $c_1, \dots, c_d$  with a common auxiliary algebra  $K$ . The corresponding  $a(c_i)$  form a basis of  $U$  by the Vandermonde determinant. It follows that

$$UV = \sum_{i=1}^d a(c_i)V = \sum_{i=1}^d T_{a(c_i)}, \quad KUV = UV.$$

The first equality with the  $T$ -spaces follows by the two inclusions in (13). Therefore  $K \subseteq \text{Stab}(UV)$ , and

$$\dim UV + \dim \text{Stab}(UV) \geq \dim T_{a(c_1)} + \dim K \geq \dim U + \dim V.$$

This proves (15).

Finally, if  $z \in \text{Stab}(W)$  is nilpotent, then  $z_{\mathbb{C}^R} = 0$  and  $\Lambda_A(z_A \pi_A(W)) = 0$ . The projection  $\pi_A(W) = A$ , since every  $U_i$  contains 1 and (14) holds. Again nondegeneracy gives  $z = 0$ , proving that  $\text{Stab}(W)$  is reduced.  $\square$

**Corollary 4.3** (Trivial-stabilizer bound). *Under the hypotheses of Lemma 4.2, if  $\text{Stab}(W) = \mathbb{C}1$ , then*

$$\dim W \geq \sum_{i=1}^m \dim U_i - (m-1). \quad (16)$$

*Proof.* Every  $\text{Stab}(W_j)$  is contained in  $\text{Stab}(W)$ , because multiplication by the remaining factors preserves its stabilizing action. Its dimension is therefore one. Sum (15) for  $j = 2, \dots, m$ .  $\square$

## 5 Lower bound for an arbitrary ordinary decomposition

**Proposition 5.1** (Exact ordinary-rank lower bound). *For the moment tensor in (1),*

$$R(H) \geq \min\{D - r + 2, (m-1)r - (m-2)s\}. \quad (17)$$

*Proof.* Let  $R = R(H)$  and take a minimum-length ordinary decomposition,

$$H(p_1, \dots, p_m) = \sum_{b=1}^R \prod_{i=1}^m \beta_{b,i}(p_i), \quad \beta_{b,i} \in (\mathbb{C}[t]_{\leq q})^* \setminus \{0\}. \quad (18)$$

Its  $R$  decomposable tensors are linearly independent. Otherwise a nonzero linear relation would eliminate one summand after rescaling the others, contradicting minimality.

Set  $B = A \times \mathbb{C}^R$  and introduce the graph subspaces

$$\tilde{U}_i = \{([p], \beta_{1,i}(p), \dots, \beta_{R,i}(p)) : p \in \mathbb{C}[t]_{\leq q}\} \subseteq B.$$

Define

$$\tilde{\Lambda}(a, c_1, \dots, c_R) = \lambda(a) - \sum_{b=1}^R c_b.$$

Equation (18) says  $\tilde{\Lambda}(\tilde{U}_1 \cdots \tilde{U}_m) = 0$ . Choose a unit  $u_i \in \tilde{U}_i$  for every  $i$ . This is possible by avoiding the finitely many hyperplanes  $p(\alpha_a) = 0$  and  $\beta_{b,i}(p) = 0$ . Now put

$$U_i = u_i^{-1} \tilde{U}_i, \quad \Lambda(x) = \tilde{\Lambda}(u_1 \cdots u_m x), \quad W = U_1 \cdots U_m. \quad (19)$$

Then  $1 \in U_i$ ,  $\Lambda(W) = 0$ , the restriction  $\Lambda_A$  is Frobenius, and each coefficient of  $\Lambda$  on a  $\mathbb{C}$  factor is nonzero. Equation (9), multiplied by units, verifies every surjectivity hypothesis in (14).

In addition,

$$\pi_{\mathbb{C}^R}(W) = \mathbb{C}^R. \quad (20)$$

Indeed, the coordinate functionals of the unnormalized product-evaluation map are the linearly independent decomposable tensors in (18). The map is therefore onto; normalization is just invertible coordinate scaling.

By Lemma 4.2,  $K = \text{Stab}(W)$  is reduced. Write its primitive idempotents as  $e_1, \dots, e_t$ , so

$$B = \prod_{j=1}^t B_j, \quad B_j = e_j B = A_j \times \mathbb{C}^{R_j}, \quad W = \bigoplus_{j=1}^t W^{(j)}, \quad W^{(j)} = e_j W.$$

Each  $A_j$  is a product of some of the local factors in (7). Write  $r_j = \dim A_j$  and let  $s_j$  count those local factors. No  $A_j$  is zero. Otherwise (20) would give  $W^{(j)} = B_j = \mathbb{C}^{R_j}$ , contradicting  $\Lambda(W^{(j)}) = 0$  and the nonzero coordinate coefficients of  $\Lambda$ . Hence

$$r_j \geq 1, \quad s_j \geq 1, \quad t \leq s, \quad \sum_j r_j = r, \quad \sum_j R_j = R. \quad (21)$$

Put  $U_i^{(j)} = e_j U_i$ . These spaces contain the identity  $e_j$  of  $B_j$ , and their product is  $W^{(j)}$ . The contextual hypotheses pass to  $B_j$ . The stabilizer of  $W^{(j)}$  inside  $B_j$  is exactly  $\mathbb{C}e_j$ : an additional stabilizer, extended by zero to the other blocks, would be an element of  $K$  not constant on its  $j$ th primitive block. By (8) and normalization by units,

$$\dim U_i^{(j)} \geq \min\{n, r_j\}.$$

Corollary 4.3 now yields

$$\dim W^{(j)} \geq m \min\{n, r_j\} - m + 1.$$

On the other hand, the nonzero functional  $\Lambda|_{B_j}$  annihilates  $W^{(j)}$ , so  $\dim W^{(j)} \leq r_j + R_j - 1$ . Therefore

$$\boxed{R_j \geq m \min\{n, r_j\} - r_j - m + 2.} \quad (22)$$

There are two possibilities. If some  $r_j \geq n$ , then

$$R \geq R_j \geq mn - r_j - m + 2 = D - r_j + 2 \geq D - r + 2.$$

If every  $r_j < n$ , sum (22) and use (21):

$$R \geq \sum_j ((m-1)r_j - (m-2)) = (m-1)r - (m-2)t \geq (m-1)r - (m-2)s.$$

In either case (17) follows.  $\square$

*Proof of Theorem 1.1.* Propositions 3.1 and 3.2 give the upper bound for  $R_{\text{sym}}(H)$ , Proposition 5.1 gives the identical lower bound for  $R(H)$ , and always  $R(H) \leq R_{\text{sym}}(H)$ . If  $I_r$  is not one-dimensional, Lemma 2.2 gives  $D = 2r - 2$ . Then  $D - r + 2 = r$  while  $(m-1)r - (m-2)s \geq r$ , so the result is independent of  $G$ .  $\square$

*Remark 5.2* (Why the mixed-root case is included). The idempotents in the argument partition whole local factors of  $A$ , not individual derivative coordinates. Thus a repeated root cannot be silently split into reduced points. The graph subspaces record arbitrary mode-dependent factors  $\beta_{b,i}$ ; none is assumed symmetric or Vandermonde. The algebra  $A$  can be larger than a mode space. Only products of all but one mode are required to surject onto  $A$ , as verified in (9).

## 6 Border ranks and a fixed Koszul identity

**Theorem 6.1** (All border ranks). *For every complex Hankel tensor,*

$$\underline{R}(H) = \underline{R}_{\text{sym}}(H) = \underline{R}_V(H) = \text{rank } C_h = r. \quad (23)$$



*Proof.* The zero tensor is immediate. For a nonzero tensor, work in the monic coordinates of Section 2 for the upper bound. Perturb  $g$  to squarefree monic polynomials  $g_\varepsilon$  of degree  $r$ . Keep the first  $r$  moments fixed and extend them through degree  $D$  by the  $g_\varepsilon$  recurrence. The resulting moments converge to the original ones, and the Vandermonde system gives a length-at-most- $r$  Vandermonde decomposition for each perturbation. This proves  $\underline{R}_V(H) \leq r$ . It remains to prove the unrestricted border-rank lower bound.

For even  $m = 2k$ , group the modes into two groups of  $k$ . For each possible index sum choose one representative multi-index in each group. The corresponding submatrix of the flattening is  $C_h$ , so its rank gives  $\underline{R}(H) \geq r$ .

Now suppose  $m = 2k + 1$ . Put

$$a = kq + 1, \quad b = \lfloor q/2 \rfloor, \quad c = \lceil q/2 \rceil, \quad p_1 = a + b, \quad p_2 = a + c.$$

Group the modes as  $k, 1, k$ . In the two outer groups, extract one coordinate for each sum  $0, \dots, kq$ . In the middle mode, extract coordinates  $0, b, q$ . These are linear maps, even when  $q = 1$  and two extracted coordinates coincide. The resulting  $a \times 3 \times a$  tensor has slices

$$M_j = (h_{u+v+j})_{0 \leq u, v < a}, \quad j = 0, b, q.$$

Its three-slice Koszul matrix is

$$\mathcal{K} = \begin{pmatrix} 0 & M_q & -M_b \\ -M_q & 0 & M_0 \\ M_b & -M_0 & 0 \end{pmatrix}. \quad (24)$$

On an arbitrary rank-one  $a \times 3 \times a$  tensor this matrix has rank at most two: it is the tensor product of a  $3 \times 3$  skew matrix of rank two and a rank-one matrix. Therefore  $\text{rank } \mathcal{K} \leq 2\underline{R}(H)$ , by linearity and closedness of matrix-rank bounds.

Here  $C_h$  is  $p_1 \times p_2$ . For  $x \in \mathbb{C}^{p_1}$  and  $y \in \mathbb{C}^{p_2}$ , define the  $(3a) \times (p_1 + p_2)$  matrix  $E$  by

$$E(x, y) = ((x_{i+b})_{i=0}^{a-1}, (-x_i + y_{i+c})_{i=0}^{a-1}, (-y_i)_{i=0}^{a-1}). \quad (25)$$

A direct block multiplication, using  $b + c = q$ , gives the identity

$$\boxed{\mathcal{K} = E \begin{pmatrix} 0 & C_h \\ -C_h^\top & 0 \end{pmatrix} E^\top.} \quad (26)$$

It holds for every moment vector, not only a generic one.

The matrix  $E$  has full column rank. If  $E(x, y) = 0$ , then  $x_i = 0$  for  $i \geq b$ , and  $y_i = 0$  for  $i < a$ . For  $i < b$ , the middle equations give  $x_i = y_{i+c} = 0$ , since  $i + c < q \leq a$ . Thus  $x = 0$ . All remaining  $y$  coordinates are then zero by the same middle equations, because  $a \geq c$ . Consequently (26) gives  $\text{rank } \mathcal{K} = 2 \text{rank } C_h = 2r$ . This proves  $\underline{R}(H) \geq r$  also for odd order, and proves (23).  $\square$

## 7 Consequences and examples

**Corollary 7.1** (Reduced support). *If  $G$  is squarefree, then all ordinary, symmetric, and Vandermonde exact and border ranks equal  $r$ .*

*Proof.* Here  $s = r$ , so Theorem 1.1 gives exact rank  $r$ . The same distinct-root recurrence gives a Vandermonde decomposition of length  $r$ , and Theorem 6.1 gives the lower bound.  $\square$

**Corollary 7.2** (A single support point). *If  $G$  has just one projective zero, then*

$$R(H) = R_{\text{sym}}(H) = \min\{D - r + 2, (m - 1)(r - 1) + 1\}.$$

*In particular, if  $h_k \neq 0$  and  $h_j = 0$  for all  $j > k$ , with  $k \leq D/2$ , then  $r = k + 1$ ,  $s = 1$ , and*

$$R(H) = R_{\text{sym}}(H) = \min\{D - k + 1, (m - 1)k + 1\}.$$

*Proof.* The first assertion substitutes  $s = 1$ . In the second,  $t^{k+1}$  gives a recurrence and the  $(k + 1) \times (k + 1)$  leading Hankel block is anti-triangular with nonzero anti-diagonal  $h_k$ . Hence its rank is  $k + 1$ , giving the stated minimal polynomial and rank.  $\square$

**Corollary 7.3** (Maximum Hankel rank). *For fixed  $m, n$ ,*

$$\max_{H \text{ Hankel}} R(H) = \max_{H \text{ Hankel}} R_{\text{sym}}(H) = (m - 1)(n - 1) + 1.$$

*Proof.* If  $r \leq n$ , the second term of (4) is at most  $(m - 1)n - (m - 2) = (m - 1)q + 1$ , since  $s \geq 1$ . If  $r \geq n$ , the first term is at most  $D - n + 2 = (m - 1)q + 1$ . The moment sequence  $h_q = 1$ ,  $h_j = 0$  for  $j \neq q$ , has  $r = n$ ,  $s = 1$  and attains the bound.  $\square$

**Example 7.4** (A mixed repeated-root case with  $r > n$ ). Take  $m = 5$ ,  $n = 3$ , and

$$L(f) = f'(0) + f(1) + f(2), \quad h_j = \mathbf{1}_{\{j=1\}} + 1 + 2^j \quad (0 \leq j \leq 10).$$

The minimal polynomial is  $g(t) = t^2(t - 1)(t - 2)$ , so  $r = 4$ ,  $s = 3$ . The formula gives

$$R(H) = R_{\text{sym}}(H) = \min\{8, 7\} = 7, \quad \underline{R}(H) = 4.$$

The upper bound is visible in the exact decomposition

$$H = \frac{1}{5} \sum_{\zeta^5=1} \zeta^{-1}(1, \zeta, 0)^{\otimes 5} + (1, 1, 1)^{\otimes 5} + (1, 2, 4)^{\otimes 5}.$$

The lower bound is Proposition 5.1; it does not assume that an ordinary decomposition has the displayed form. This example distinguishes exact rank, border rank, and the degree-eight binary interpolation bound.

**Example 7.5** (Two repeated support points above the mode dimension). Take  $m = 5$ ,  $n = 3$ , and  $L(f) = f'(0) + f'(1)$ . Then  $g = t^2(t - 1)^2$ ,  $r = 4$ ,  $s = 2$ . The local construction gives ten terms, whereas the binary construction gives eight. Theorem 1.1 proves that both exact ranks are eight.

**Example 7.6** (Rank substantially below Vandermonde rank). For  $m \geq 3$ ,  $n = 3$ , let  $h_1 = h_{2m} = 1$  and all other moments vanish. The associated form in three tensor variables is  $mx^{m-1}y + z^m$ . Its projective apolar support has one double point and one simple point:  $r = 3$ ,  $s = 2$ . Thus

$$R(H) = R_{\text{sym}}(H) = m + 1, \quad \underline{R}(H) = 3.$$

Its binary Vandermonde rank is  $2m - 1$ , by Proposition 3.3. Equality of ordinary and symmetric rank therefore does not mean that an optimal decomposition must stay on the rational normal curve.

**Example 7.7** (A table of exact values). Each row specifies a Frobenius moment functional by the multiplicities of its minimal apolar support. The balanced row is independent of that choice.

| $m$ | $n$ | $r$ | $s$ | $D - r + 2$ | $(m - 1)r - (m - 2)s$ | $R = R_{\text{sym}}$ |
|-----|-----|-----|-----|-------------|-----------------------|----------------------|
| 3   | 3   | 3   | 1   | 5           | 5                     | 5                    |
| 3   | 4   | 5   | 4   | 6           | 6                     | 6                    |
| 4   | 3   | 4   | 2   | 6           | 8                     | 6                    |
| 5   | 3   | 4   | 3   | 8           | 7                     | 7                    |
| 5   | 3   | 4   | 2   | 8           | 10                    | 8                    |
| 5   | 3   | 6   | 1   | 6           | 21                    | 6                    |
| 7   | 4   | 5   | 4   | 18          | 10                    | 10                   |

## 8 An exact rank algorithm

For exact rational or algebraic input, no root approximation is required. Form the matrices

$$C_d = (h_{i+j})_{\substack{0 \leq j \leq D-d \\ 0 \leq i \leq d}} \quad (d = 1, \dots, \lfloor D/2 \rfloor + 1).$$

The first matrix with nonzero kernel gives  $r$  and coefficients  $(g_0, \dots, g_r)$  of  $G = \sum g_i X^{r-i} Y^i$ . Let  $g(t) = \sum g_i t^i$ , and let  $e = \deg g$  in the original coordinates. The number of distinct projective zeros is

$$s = e - \deg \gcd(g, g') + \mathbf{1}_{\{e < r\}}. \quad (27)$$

For a nonzero constant  $g$ , the first two terms are zero. The final indicator counts the possible root at infinity only once, irrespective of its multiplicity. Return the minimum in (4).

The supplied `code/hankel_rank.py` implements these operations in SymPy using exact rational arithmetic. It reports the two upper bounds, the apolar polynomial, the ordinary and symmetric ranks, and the common border rank. It rejects floating-point input rather than deciding exact rank with a numerical tolerance. Rank computation does not require factoring  $G$  into linear factors.

The companion `code/binary_certificate.py` constructs a minimal Vandermonde decomposition implicitly, without approximating its algebraic roots. Its output consists of a squarefree monic polynomial  $P$ , a polynomial  $U$ , and a vector of polynomials  $f$ , all over  $\mathbb{Q}$ , with

$$H = \sum_{P(\beta)=0} \frac{U(\beta)}{P'(\beta)} f(\beta)^{\otimes m}.$$

The top-coefficient identity for Lagrange interpolation verifies this sum by polynomial remainders in  $\mathbb{Q}[t]/(P)$ . A rational projective shear handles roots initially at infinity. These Vandermonde certificates are optimal for ordinary and symmetric rank precisely when their length agrees with (4); the output explicitly labels this distinction.

An optimal symmetric decomposition can also be constructed. If the second term of (4) is smaller, use the Chinese remainder and root-of-unity construction in Proposition 3.1. If the first term is smaller, use the squarefree apolar pencil in Proposition 3.2. Any zero or redundant summands would contradict the lower bound, so a length-achieving construction is minimal.

## 9 Verification scope

The mathematical proof of the universal statement is Sections 2–5. The finite computations supplied with this manuscript are reproducibility checks, not a replacement for that proof. They check exact catalecticant and apolar calculations, local Fourier interpolation identities, complete small-dimensional tensor reconstructions, a mixed-root graph-algebra example, and the fixed identity (26) across many dimensions and moment patterns. Finite-field checks are labeled as such and do not purport to prove the complex-field theorem by testing.

A separate Codex AI agent independently audited this complete argument on 12 September 2026 and returned PASS under the repository’s informal-audit policy. The dated review is available at <https://github.com/sidneyholden1/OpenProblemsInNLA/blob/codex/holden-tr14-resolution/references/holden-tr14-2026-09-12/independent-review.md>.

AI assistance was used for review and submission preparation. This does not constitute external human peer review or formal verification; no Lean checking was performed. In particular, a successful run of the verification scripts is not a certificate for an arbitrary claimed mathematical proof. The companion `PROOF_AUDIT.md` identifies the dependencies and checks the surjectivity, nilpotent exclusion, and support-partition steps explicitly.

## A An independent lower-bound check for terminal moment sequences

This appendix provides an elementary verification of the single-support formula without the contextual product-space argument.

**Lemma A.1** (Terminal-sequence substitution bound). *Let  $m \geq 2$ ,  $q \geq 1$ ,  $q \leq k \leq mq$ , and let  $T$  have entries  $h_{i_1+\dots+i_m}$ ,  $0 \leq i_j \leq q$ , where  $h_k \neq 0$  and  $h_j = 0$  for  $j > k$ . Then  $R(T) \geq mq - k + 1$ .*

*Proof.* Induct on  $m$ . For  $m = 2$ , an anti-triangular submatrix of size  $2q - k + 1$  has nonzero anti-diagonal  $h_k$ . For general  $m$ , if  $k > (m - 1)q$ , the first-mode flattening contains an anti-triangular minor of size  $mq - k + 1$ : use row indices  $i = k - (m - 1)q, \dots, q$  and column sums  $k - q, \dots, (m - 1)q$ . It remains that  $q \leq k \leq (m - 1)q$ . The first-mode flattening has full row rank  $q + 1$ , using rows  $0, \dots, q$  and column sums  $k - q, \dots, k$ . In a minimum decomposition, choose  $q + 1$  independent first-mode factors and their dual basis  $p_0, \dots, p_q$  in  $\mathbb{C}[t]_{\leq q}$ . At least one  $p_a$  has  $p_a(0) \neq 0$ . Contraction with that polynomial kills  $q$  of the chosen summands and gives an  $(m - 1)$ -mode Hankel tensor with moments  $h'_j = \sum_{u=0}^q (p_a)_u h_{j+u}$ . It has  $h'_k = (p_a)_0 h_k \neq 0$  and  $h'_j = 0$  for  $j > k$ . Induction gives its rank at least  $(m - 1)q - k + 1$ ; restoring the  $q$  killed summands gives  $mq - k + 1$ .  $\square$

For  $1 \leq k \leq q$ , all entries with a mode index above  $k$  vanish. Restrict each mode to indices  $0, \dots, k$  and apply Lemma A.1 with  $q' = k$ ; this gives  $(m - 1)k + 1$ . The case  $k = 0$  has rank one directly. For  $k \geq q$ , the lemma supplies  $D - k + 1$ . When  $k \leq D/2$ , these lower bounds match the two branches of the single-support formula and the symmetric upper bounds in Section 3. Thus no contextual product inequality is needed for this check, including the extremal example in Corollary 7.3.

## B Source notes

The contextual inequality in Lemma 4.2, and its use with the graph of an unrestricted decomposition, are the essential arguments of this manuscript. The linear transformation in Lemma 4.1 is credited to Beck–Lecouvey. Its finiteness conclusion here is not assumed from a theorem requiring finitely many subalgebras of the ambient nonreduced algebra; it is proved by excluding nilpotents from each auxiliary algebra.

For  $r \leq n$ , the lower bound reduces to the familiar commutative Alder–Strassen-type expression. The general multilinear algebra lower bound provides a useful comparison, but is not used to pass a rank bound through a restriction.<sup>5</sup>

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<sup>5</sup>M. Bläser, H. Mayer, and D. Shringi, *On the Multilinear Complexity of Associative Algebras*, STACS 2023, Article 12, Theorem 19. <https://doi.org/10.4230/LIPIcs.STACS.2023.12>.